Capstone Checkpoint 4.1 Graph Retriver Evaluation

Oracle, vector, and hybrid experiment comparison. Generated 2026-09-15 19:17 EDT.

14	4	266	19	0
Included runs	Configurations	Question-runs	Questions	Legacy runs excluded

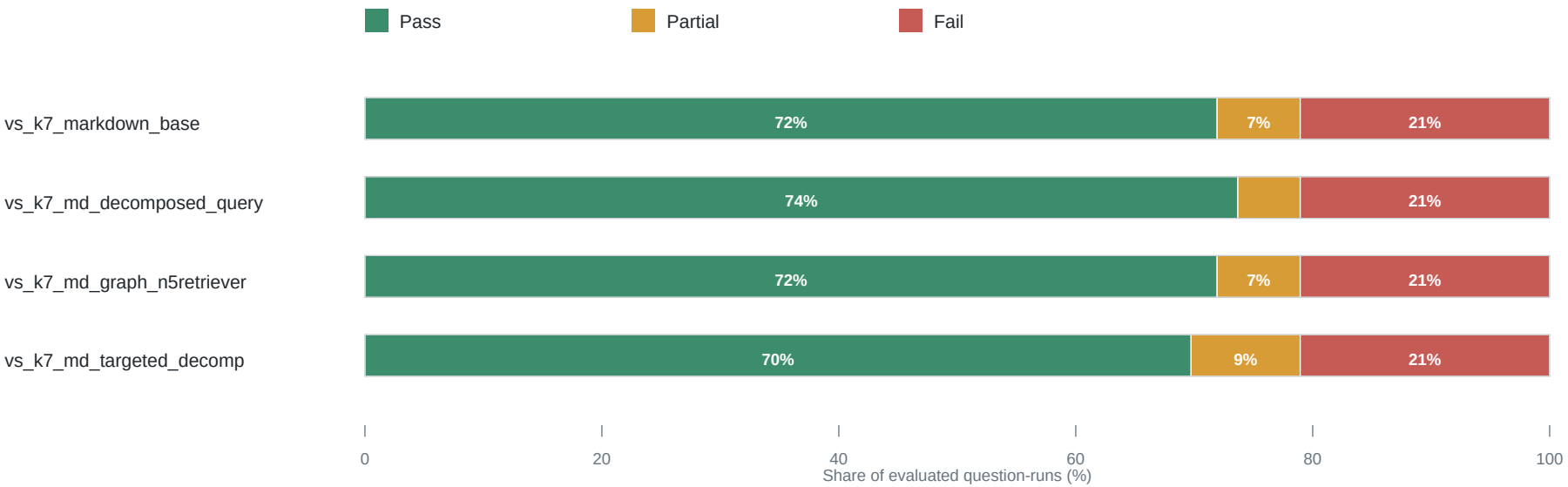
Summary

The highest observed mean quality score was 0.763 for vs_k7_md_decomposed_query across 4 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 93.9% for vs_k7_md_graph_n5retriever. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 62.5 for Exhaustive list or time filter under vs_k7_md_decomposed_query. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

Answer outcomes



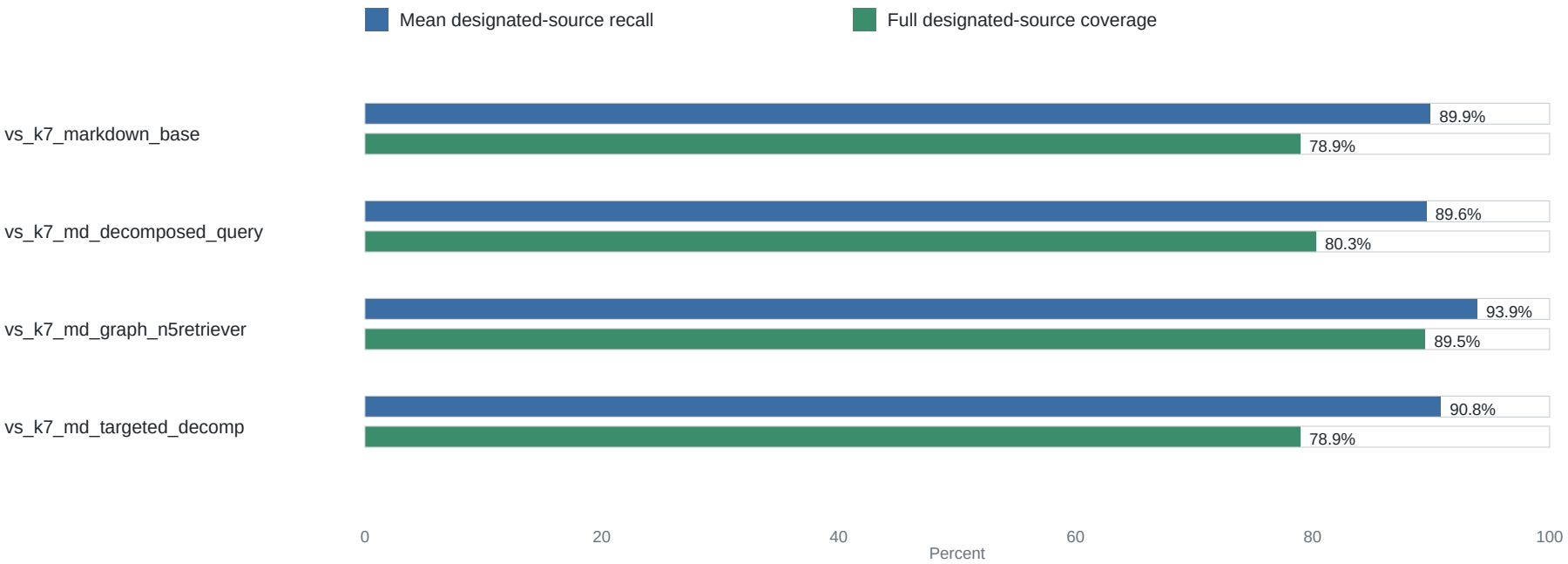
Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
vs_k7_markdown_base	3	57	71.9%	7.0%	21.1%	0.754	0.033	89.9%	78.9%	207.1k
vs_k7_md_decomposed_query	4	76	73.7%	5.3%	21.1%	0.763	0.037	89.6%	80.3%	240.5k
vs_k7_md_graph_n5retriever	3	57	71.9%	7.0%	21.1%	0.754	0.045	93.9%	89.5%	260.7k

Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
vs_k7_md_targeted_decomp	4	76	69.7%	9.2%	21.1%	0.743	0.029	90.8%	78.9%	245.3k

Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.

Retrieval diagnostics

Designated-source coverage



The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.

Outcome and source-coverage combinations

Configuration	Pass, full	Pass, incomplete	Non-pass, full	Non-pass, missing
vs_k7_markdown_base	35	6	10	6
vs_k7_md_decomposed_query	47	9	14	6
vs_k7_md_graph_n5retriever	38	3	13	3
vs_k7_md_targeted_decomp	44	9	16	7

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

	vs_k7_markdown_base	vs_k7_md_decomposed_query	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
GT01	**1.00**	**1.00**	**1.00**	**1.00**
GT02	**0.83**	**0.88**	**1.00**	**1.00**
GT03	**0.00**	**0.00**	**0.00**	**0.00**
GT04	**1.00**	**1.00**	**1.00**	**1.00**
GT05	**1.00**	**1.00**	**1.00**	**1.00**
GT06	**1.00**	**1.00**	**1.00**	**1.00**
GT07	**1.00**	**0.88**	**0.33**	**0.50**
GT08	**0.33**	**0.38**	**0.67**	**0.25**
GT09	**1.00**	**1.00**	**1.00**	**1.00**
GT10	**0.00**	**0.00**	**0.00**	**0.00**
GT11	**0.00**	**0.00**	**0.00**	**0.12**
GT12	**1.00**	**1.00**	**1.00**	**0.88**
GT13	**0.83**	**1.00**	**1.00**	**0.88**
GT14	**1.00**	**1.00**	**1.00**	**1.00**
GT15	**1.00**	**1.00**	**1.00**	**1.00**
GT16	**0.33**	**0.38**	**0.50**	**0.50**
GT17	**1.00**	**1.00**	**1.00**	**1.00**
GT18	**1.00**	**1.00**	**1.00**	**1.00**
GT19	**1.00**	**1.00**	**0.83**	**1.00**

Fail (0) Partial (0.5) Pass (1)

Question	vs_k7_markdown_base	vs_k7_md_decomposed_query	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
GT01	1.000	1.000	1.000	1.000
GT02	0.833	0.875	1.000	1.000
GT03	0.000	0.000	0.000	0.000
GT04	1.000	1.000	1.000	1.000
GT05	1.000	1.000	1.000	1.000
GT06	1.000	1.000	1.000	1.000
GT07	1.000	0.875	0.333	0.500
GT08	0.333	0.375	0.667	0.250
GT09	1.000	1.000	1.000	1.000
GT10	0.000	0.000	0.000	0.000
GT11	0.000	0.000	0.000	0.125
GT12	1.000	1.000	1.000	0.875
GT13	0.833	1.000	1.000	0.875
GT14	1.000	1.000	1.000	1.000
GT15	1.000	1.000	1.000	1.000
GT16	0.333	0.375	0.500	0.500
GT17	1.000	1.000	1.000	1.000
GT18	1.000	1.000	1.000	1.000
GT19	1.000	1.000	0.833	1.000

Per-question article recall

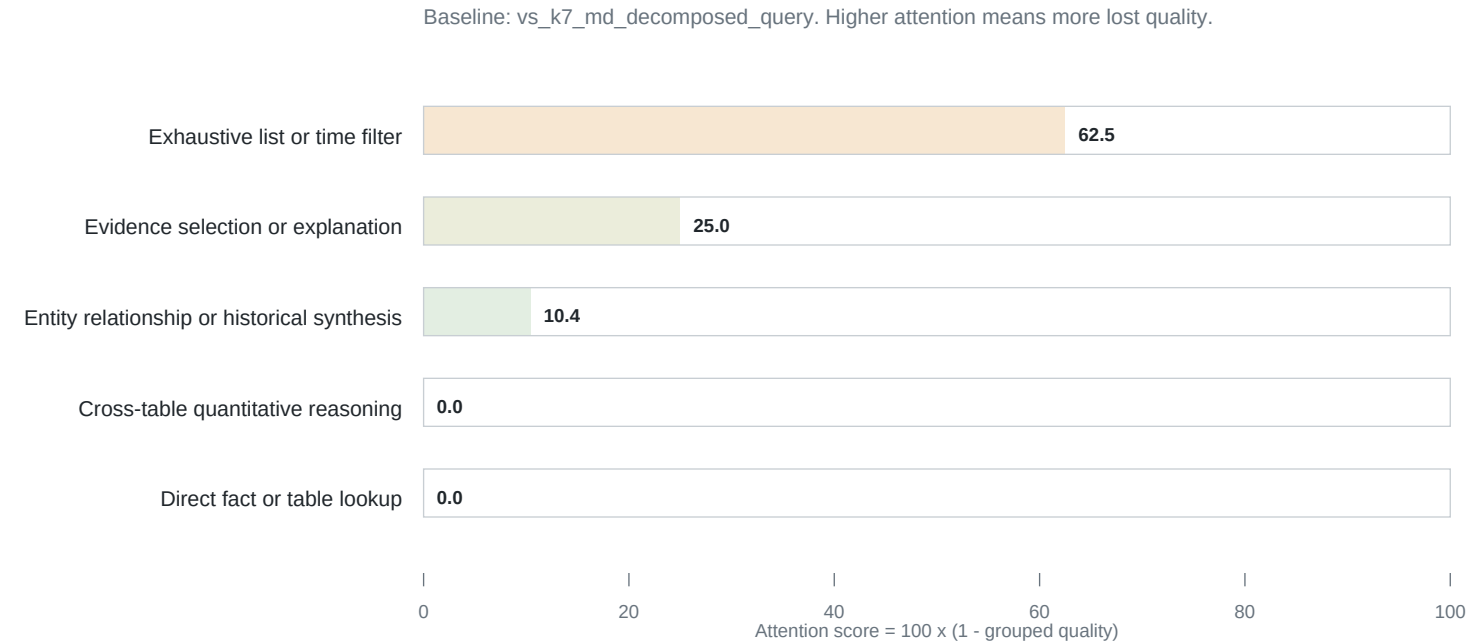
Fraction of each question's designated source articles that appeared in the retrieved set (retriever configurations only). 1.00 = every gold article retrieved; lower values mark a retrieval miss — usually a multi-source question.

Question	vs_k7_markdown_base	vs_k7_md_decomposed_query	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
GT01	100.0%	100.0%	100.0%	100.0%

Question	vs_k7_markdown_base	vs_k7_md_decomposed_query	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
GT02	100.0%	100.0%	100.0%	100.0%
GT03	33.3%	16.7%	33.3%	50.0%
GT04	100.0%	100.0%	100.0%	100.0%
GT05	100.0%	100.0%	100.0%	100.0%
GT06	100.0%	100.0%	100.0%	100.0%
GT07	100.0%	100.0%	100.0%	100.0%
GT08	100.0%	100.0%	100.0%	100.0%
GT09	50.0%	100.0%	50.0%	100.0%
GT10	100.0%	100.0%	100.0%	100.0%
GT11	100.0%	100.0%	100.0%	100.0%
GT12	100.0%	91.7%	100.0%	75.0%
GT13	100.0%	100.0%	100.0%	100.0%
GT14	75.0%	81.2%	100.0%	87.5%
GT15	100.0%	100.0%	100.0%	87.5%
GT16	50.0%	62.5%	100.0%	75.0%
GT17	100.0%	50.0%	100.0%	50.0%
GT18	100.0%	100.0%	100.0%	100.0%
GT19	100.0%	100.0%	100.0%	100.0%

Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.



Task family	Questions	Baseline quality	Attention	Fail	Oracle quality	Oracle gap	Article recall
Exhaustive list or time filter	5	0.375	62.5	60.0%	n/a	n/a	83.3%
Evidence selection or explanation	3	0.750	25.0	16.7%	n/a	n/a	100.0%
Entity relationship or historical synthesis	6	0.896	10.4	8.3%	n/a	n/a	80.9%
Cross-table quantitative reasoning	2	1.000	0.0	0.0%	n/a	n/a	100.0%
Direct fact or table lookup	3	1.000	0.0	0.0%	n/a	n/a	100.0%

Baseline quality and failure rate use vs_k7_md_decomposed_query. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.

Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

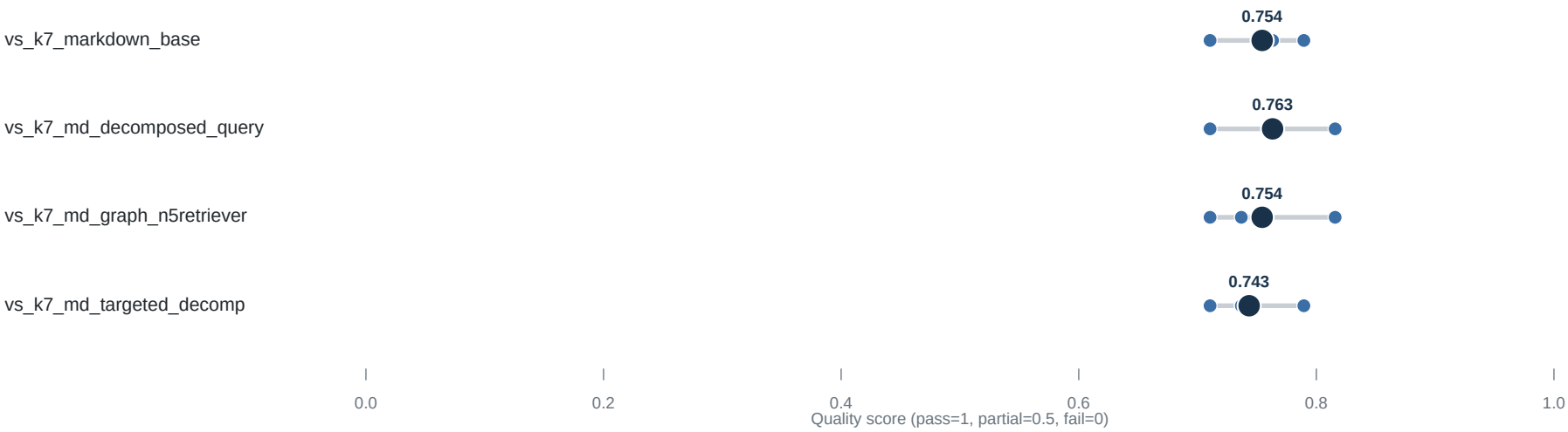
	vs_k7_markdown_base	vs_k7_md_decomposed_query	vs_k7_md_graph_n5retriever	vs_k7_md_targeted_decomp
1 article	**0.727**	**0.739**	**0.727**	**0.705**
2 articles	**0.867**	**0.875**	**0.867**	**0.900**
3+ articles	**0.667**	**0.667**	**0.667**	**0.625**

Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	vs_k7_markdown_base	11	33	69.7%	6.1%	24.2%	0.727	100.0%
2 articles	vs_k7_markdown_base	5	15	80.0%	13.3%	6.7%	0.867	80.0%
3+ articles	vs_k7_markdown_base	3	9	66.7%	0.0%	33.3%	0.667	69.4%
1 article	vs_k7_md_decomposed_query	11	44	70.5%	6.8%	22.7%	0.739	100.0%
2 articles	vs_k7_md_decomposed_query	5	20	85.0%	5.0%	10.0%	0.875	82.5%
3+ articles	vs_k7_md_decomposed_query	3	12	66.7%	0.0%	33.3%	0.667	63.2%
1 article	vs_k7_md_graph_n5retriever	11	33	69.7%	6.1%	24.2%	0.727	100.0%
2 articles	vs_k7_md_graph_n5retriever	5	15	80.0%	13.3%	6.7%	0.867	90.0%
3+ articles	vs_k7_md_graph_n5retriever	3	9	66.7%	0.0%	33.3%	0.667	77.8%
1 article	vs_k7_md_targeted_decomp	11	44	65.9%	9.1%	25.0%	0.705	100.0%
2 articles	vs_k7_md_targeted_decomp	5	20	85.0%	10.0%	5.0%	0.900	82.5%
3+ articles	vs_k7_md_targeted_decomp	3	12	58.3%	8.3%	33.3%	0.625	70.8%

Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.

Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.



Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
zealous_ritchie	vs_k7_markdown_base	19	14	1	4	0.763	89.9%	78.9%
relaxed_musk	vs_k7_markdown_base	19	12	3	4	0.711	89.9%	78.9%
elastic_adleman	vs_k7_markdown_base	19	15	0	4	0.789	89.9%	78.9%
practical_adleman	vs_k7_md_decomposed_query	19	15	1	3	0.816	88.2%	78.9%
pensive_shannon	vs_k7_md_decomposed_query	19	13	1	5	0.711	93.9%	89.5%
admiring_matsumoto	vs_k7_md_decomposed_query	19	14	1	4	0.763	88.2%	78.9%
amazing_gosling	vs_k7_md_decomposed_query	19	14	1	4	0.763	88.2%	73.7%
bold_wozniak	vs_k7_md_graph_n5retriever	19	13	2	4	0.737	93.9%	89.5%
awesome_valiant	vs_k7_md_graph_n5retriever	19	13	1	5	0.711	93.9%	89.5%
reverent_wilkes	vs_k7_md_graph_n5retriever	19	15	1	3	0.816	93.9%	89.5%
relaxed_ullman	vs_k7_md_targeted_decomp	19	14	2	3	0.789	92.1%	84.2%
eager_wozniak	vs_k7_md_targeted_decomp	19	13	1	5	0.711	90.8%	78.9%

Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
pedantic_boole	vs_k7_md_targeted_decomp	19	13	2	4	0.737	89.9%	73.7%
ecstatic_tanenbaum	vs_k7_md_targeted_decomp	19	13	2	4	0.737	90.4%	78.9%

Methodology and limitations

Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

Configuration grouping

Runs are grouped by a CSV experiment_name or config_name when one is present. If no explicit name is present, grouping falls back to evaluation mode, context source, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

Included experiment files

Run file
experiments/admiring_matsumoto.csv
experiments/amazing_gosling.csv
experiments/awesome_valiant.csv
experiments/bold_wozniak.csv
experiments/eager_wozniak.csv
experiments/ecstatic_tanenbaum.csv
experiments/elastic_adleman.csv
experiments/pedantic_boole.csv
experiments/pensive_shannon.csv
experiments/practical_adleman.csv
experiments/relaxed_musk.csv

Run file
experiments/relaxed_ullman.csv
experiments/reverent_wilkes.csv
experiments/zealous_ritchie.csv

Excluded files

Run file	Reason
datasets/wikipedia_GT_advanced_inputs.csv	missing columns: ['score']
datasets/wikipedia_GT_graph_inputs.csv	duplicate of datasets/wikipedia_GT_advanced_inputs.csv
datasets/wikipedia_GT_oracle_inputs.csv	duplicate of datasets/wikipedia_GT_advanced_inputs.csv
datasets/wikipedia_GT_retrieval_inputs.csv	duplicate of datasets/wikipedia_GT_advanced_inputs.csv